

```
sysadmin@localhost:~$ vi myfile
```

To create a new file, execute the following command:

```
#!/bin/bash
echo "Hello there! Here is the calendar for this month:"
cal
```

In Vi mode - clear that this is a BASH shell script, you need to include a special line at the top of the file called a "shbang". This line starts with `#!` and then contains the path to the BASH shell executable. Add the following line at the top of the `sample.sh` file

```
sysadmin@localhost:~$ bash sample.sh
Hello there! Here is the calendar for this month:
    December 2018
Su Mo Tu We Th Fr Sa
                1
 2  3  4  5  6  7  8
 9 10 11 12 13 14 15
16 17 18 19 20 21 22
23 24 25 26 27 28 29
30 31

sysadmin@localhost:~$
```

Using `bash sample.sh` to execute the script

```
sysadmin@localhost:~$ vi sample.sh
sysadmin@localhost:~$ ls -l sample.sh
-rw-rw-r-- 1 sysadmin sysadmin 74 Oct 10 07:31 sample.sh
sysadmin@localhost:~$ chmod a+x sample.sh
sysadmin@localhost:~$ ./sample.sh
Hello there! Here is the calendar for this month:
    October 2026
Su Mo Tu We Th Fr Sa
      1  2  3
 4  5  6  7  8  9 10
11 12 13 14 15 16 17
18 19 20 21 22 23 24
25 26 27 28 29 30 31
```

`chmod` command is used to change permissions on the file so that the file can be executed.

```
sysadmin@localhost:~$ sample.sh
sample.sh: command not found
sysadmin@localhost:~$ echo $PATH
/home/sysadmin/bin:/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/snap/bin
sysadmin@localhost:~$ mkdir bin
sysadmin@localhost:~$ mv sample.sh bin
sysadmin@localhost:~$ sample.sh
Hello there! Here is the calendar for this month
    October 2026
Su Mo Tu We Th Fr Sa
      1  2  3
 4  5  6  7  8  9 10
11 12 13 14 15 16 17
18 19 20 21 22 23 24
25 26 27 28 29 30 31
```

```
#!/bin/bash
echo "Please enter your age"
read age
if test $age -lt 16
then
echo "You are not old enough to drive."
else
echo "You can drive!"
fi
~
```

Create the following `drive.sh` file and make it executable to see how the `if` and `test` statements work

```
sysadmin@localhost:~$ vi drive.sh

[8]+  Stopped                  vi drive.sh
sysadmin@localhost:~$ chmod a+x drive.sh
sysadmin@localhost:~$ ./drive.sh
Please enter your age
23
You can drive!
sysadmin@localhost:~$ ./drive.sh
Please enter your age
6
You are not old enough to drive.
sysadmin@localhost:~$ vi drive.sh

[9]+  Stopped                  vi drive.sh
sysadmin@localhost:~$
```

you could read the `if` statement as If `$age` is less than 16, then `echo` 'You are not old enough to drive', else `echo` 'You can drive!'. The `fi` ends the `if` statement.

```

sysadmin@localhost:~$ cat check.sh
#!/bin/bash
echo "Enter a username to check: "
read name
if grep $name /etc/passwd > /dev/null
then
    echo "$name is on this system"
else
    echo "$name does not exist"
fi
sysadmin@localhost:~$ chmod a+x check.sh
sysadmin@localhost:~$ ./check.sh
Enter a username to check:
root
root is on this system
sysadmin@localhost:~$ ./check.sh
Enter a username to check:
bobby
bobby does not exist
sysadmin@localhost:~$

```

number the third time. Your screen should look like the following.

```

sysadmin@localhost:~$ cat num.sh
#!/bin/bash
echo "Please enter a number greater than 100"
read num
while [ $num -le 100 ]
do
    echo "$num is NOT greater than 100."
    echo "Please enter a number greater than 100."
    read num
done
echo "Finally, $num is greater than 100"
sysadmin@localhost:~$ chmod a+x num.sh
sysadmin@localhost:~$ ./num.sh
Please enter a number greater than 100
25
25 is NOT greater than 100.
Please enter a number greater than 100.
99
99 is NOT greater than 100.
Please enter a number greater than 100.
101
Finally, 101 is greater than 100
sysadmin@localhost:~$

```